

Literature-Already-Answered Status Note

Bornological differentiability and weakly compact operators

Status: weak-Hadamard case and property-(S) framework answered in later literature

Source question

Bachir, Flores, and Tapia-García ask in Open Question 5.7 of *Compact and Limited Operators* (arXiv:1910.07362, PDF page 13) whether there is a differentiability characterization for the weak-Hadamard bornology and, more generally, for a prescribed bornology β on a Banach space [1]. In their notation, they seek a class $\mathcal{F}_\beta$ such that

$$T(B_Y) \in \beta \iff f \circ T \text{ is Fréchet differentiable at } y$$

whenever $f \in \mathcal{F}_\beta$ is Gâteaux differentiable at Ty .

Explicit later answer

Bachir and Tapia-García explicitly state on PDF page 2 of *Non-linear operators and differentiability of Lipschitz functions* (arXiv:2110.00099) that their work answers Question 5.7 of the source [2]. Their Theorem 1.3 (PDF page 3) gives the highlighted missing case. A bounded linear map $T : Y \rightarrow X$ is weakly compact if and only if, for every Lipschitz $f : X \rightarrow \mathbb{R}$, the composition $f \circ T$ is Fréchet differentiable at y whenever f is weak-Hadamard differentiable at Ty . Thus the desired class is the real-valued Lipschitz functions, with weak-Hadamard differentiability imposed at the image point.

The same paper also gives the intended abstract framework. Theorem 4.5 (PDF page 11, proof through page 13) treats vector bornologies β_X and β_Y when β_X has property (S). It characterizes when the positively homogeneous differential of T maps β_Y -sets to β_X -sets by the composition behavior of all real-valued Lipschitz functions that are β_X -differentiable. The paper proves property (S) for the Hadamard, weak-Hadamard, and limited bornologies, thereby recovering compact and limited operators and adding weakly compact and completely continuous operators.

Classification and scope

The match is explicit: the supporting authors cite the source as reference [4], name Question 5.7, and say their paper answers it. The weak-Hadamard case is completely settled by Theorem 1.3, so it is not a viable target for a new proof. The abstract theorem is broad but assumes a vector bornology satisfying property (S). Consequently, it should not be read as an assertion for every possible bornology with no structural hypothesis; that unrestricted literal formulation lies outside the theorem.

The status search used the source arXiv id, the exact phrase “weak-Hadamard differentiability,” and combinations of “bornology,” “operator,” and “differentiability.” It located arXiv:2110.00099, whose abstract, introduction, Theorem 1.3, and Theorem 4.5 provide the decisive identification.

References

- [1] M. Bachir, G. Flores, and S. Tapia-García, *Compact and Limited Operators*, Math. Nachr. 294 (2021), 1085–1098, [arXiv:1910.07362](#).
- [2] M. Bachir and S. Tapia-García, *Non-linear operators and differentiability of Lipschitz functions*, [arXiv:2110.00099](#), 2021.